Beam Resource Matching for Pilot Assignment

Combining Beam-Domain Conflict Modeling with Many-to-Many Matching

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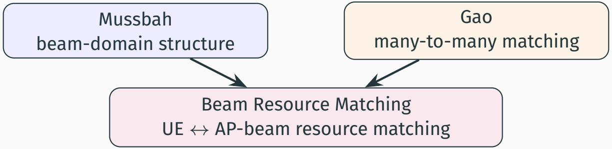
Core Design Concept

Design principle

Beam Resource Matching lifts Gao's many-to-many matching from the AP domain to the AP-beam resource domain and inherits Mussbah's active/moderate beam structure.



- **Gao**: many-to-many matching between UEs and APs on large-scale fading.
- **Mussbah**: beam-domain conflict graph from active/moderate beam overlaps.
- **Beam Resource Matching**: matches UEs to resources $r = (l, n)$; the matching outcome determines the active beam set and feeds pilot assignment.

Main Object: AP-Beam Resource

Resource definition

$$r = (l, n), \qquad |\mathcal{R}| = LN.$$

- l: AP index.
- n: beam index at AP l.
- Each pair (l, n) constitutes a single matching resource.

UE-resource preference

$$P_{k,r} = P_{k,l,n}.$$

Resolution gain over Gao

- Gao ranks candidates by AP-domain strength $\beta_{k,l}$.
- Beam Resource Matching ranks by beam-domain power $P_{k,l,n}$.
- Beam-level resolution separates "same AP" from "same beam."

Modeling distinction

Matching operates at the AP-beam resource level rather than the AP level.

Mussbah Beam-Domain Method

Beam-domain channel

$$\mathbf{h}_{k,l}^{(B)} = \mathbf{U}^H \mathbf{h}_{k,l}.$$

UE k reports beam set $\mathcal{B}_k$. The strongest beams (cumulative power threshold) form the active set:

$$\mathcal{B}_k^{(a)} \subset \mathcal{B}_k.$$

Remaining reported beams form the moderate set:

$$\mathcal{B}_k^{(i)} = \mathcal{B}_k \setminus \mathcal{B}_k^{(a)}.$$

Conflict graph

$$\mathbf{B} = \mathbf{B}^{(a)\,T}\mathbf{B}^{(a)} + \mathbf{B}^{(a)\,T}\mathbf{B}^{(i)} + \mathbf{B}^{(i)\,T}\mathbf{B}^{(a)}.$$

$$A_{i,j} = 1 \quad \text{if } B_{i,j} > 0.$$

Pilots are assigned by graph coloring:

$$\tau_{\mathrm{p}}^{\mathrm{actual}} = \chi(G).$$

Beam-domain conflicts capture finer interference than AP-domain. Trade-off: dense graphs raise the chromatic number and inflate pilot overhead.

Design objective

Beam Weighted Threshold retains Mussbah's beam-domain coloring framework while suppressing weak edges through a weighted interference score.

$$B^{w}_{u,v} = w_{\mathrm{aa}}\,|\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(a)}| + w_{\mathrm{ai}}\,|\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(i)}| + w_{\mathrm{ia}}\,|\mathcal{B}_u^{(i)} \cap \mathcal{B}_v^{(a)}|.$$

$$A_{u,v} = \begin{cases} 1, & B^{w}_{u,v} > \theta,\ u \neq v, \\ 0, & \text{otherwise.} \end{cases}$$

- w_{aa} emphasizes active-active overlap, the typically dominant interference.
- Threshold θ controls graph density.
- Pilots are assigned by graph coloring on G_θ:

$$\tau_{\mathrm{p}}^{\mathrm{actual}} = \chi(G_\theta).$$

Gao AP-Domain Matching

Gao performs many-to-many matching between UEs and APs based on AP-domain large-scale fading $\beta_{m,k}$. The indicator

$$\mu(m, k) = 1$$

denotes that UE m is matched to AP k.

Quota constraints

$$|\mu(\mathrm{UE}_m)| \leq Q_{\mathrm{UE}}, \qquad |\mu(\mathrm{AP}_k)| \leq Q_{\mathrm{AP}}.$$

Key setting:

$$Q_{\mathrm{AP}} = \tau_{\mathrm{p}}.$$

AP group

$$\mathcal{G}_k = \{m : \mu(m, k) = 1\}.$$

Common serving AP ratio

$$M^{AP}_{m_1,m_2} = \frac{|\mu(m_1,:) \cap \mu(m_2,:)|}{|\mu(m_1,:)|}.$$

Group risk

$$S_k = \sum_{m_1 \neq m_2,\ m_1,m_2 \in \mathcal{G}_k} M^{AP}_{m_1,m_2}.$$

Limitation: AP-level matching cannot distinguish two UEs that share an AP through different beams.

AP-Top-N Method

Sparse AP-domain conflict graph

$$\mathcal{T}_k(N_{\text{top}}) = \text{Top}_{N_{\text{top}}}\{\beta_{k,1}, \dots, \beta_{k,L}\}.$$

A conflict edge is placed when two UEs share at least one Top-N AP:

$$e_{i,j}^{\text{TopN}} = \mathbf{1}\{\mathcal{T}_i(N_{\text{top}}) \cap \mathcal{T}_j(N_{\text{top}}) \neq \varnothing\}.$$

Pilot assignment flow

- Identify the Top-N strongest APs per UE.
- Connect UEs whose Top-N sets intersect.
- Apply graph coloring on the resulting sparse graph.
- Connected UEs receive distinct pilots.

Algorithmic role

AP-Top-N provides a lightweight AP-domain sparsification of Gao's AP-overlap criterion.

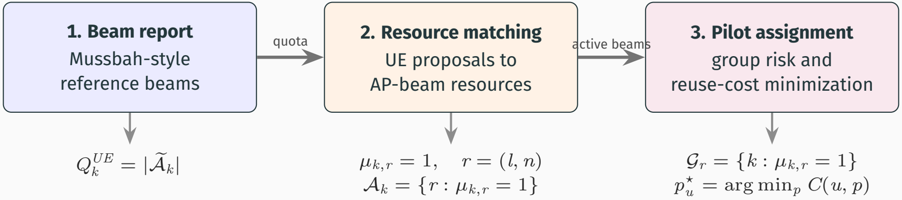
Method Comparison Before BRM

Method	Basis	Assignment principle
Gao	AP-domain matching	Match UEs to APs and compute AP group risk
Beam Weighted Threshold	beam-domain	Create edges only when weighted beam overlap exceeds a threshold
Beam Resource Matching	AP-beam resource	Match UEs to AP-beam resources
AP-Top-N	AP-domain sparsification	Use only strongest AP Top-N overlap as conflict

Methodological position

AP-Top-N reuses Gao's large-scale AP information; Beam Resource Matching combines Mussbah's beam-domain structure with Gao's matching mechanism.

BRM: Overall Flow



Operational mechanism

Beam reports set the UE quotas; matching determines the final active beam set; weighted overlap drives pilot assignment.

BRM: Matching Variables and Constraints

Reference quota

$$Q_k^{UE} = |\widetilde{\mathcal{A}}_k|.$$

$\widetilde{\mathcal{A}}_k$ denotes the reference active beam set from the Mussbah power-threshold rule.

Matching variable

$$\mu_{k,r} \in \{0, 1\}.$$

$$\mu_{k,r} = 1 \quad \Longleftrightarrow \quad \text{UE } k \text{ is matched to resource } r.$$

Constraints

$$\sum_{r \in \mathcal{R}} \mu_{k,r} \leq Q_k^{UE}, \qquad \sum_{k=1}^{K} \mu_{k,r} \leq Q_{\text{res}}.$$

$$\mu_{k,r} = 0 \quad \text{if } r \notin \mathcal{B}_k.$$

Typical choice:

$$Q_{\text{res}} = \tau_{\text{p}}^{\text{design}}.$$

Matching follows a deferred-acceptance protocol: UEs propose to their strongest resources, and each resource accepts up to Q_{res} UEs ranked by beam power.

Matching Result Defines Active Beams

Mussbah selects active beams via a per-UE power threshold.

Beam Resource Matching defines active beams as the matched resources:

$$\mathcal{A}_k = \{r : \mu_{k,r} = 1\}.$$

Unmatched reported beams form the moderate set:

$$\mathcal{I}_k = \mathcal{B}_k \setminus \mathcal{A}_k.$$

Algorithmic consequence

Active beams reflect global UE-resource competition, not only per-UE power ranking.

- Each beam must survive UE-resource competition.
- Resource quota Q_{res} caps UE concentration per AP-beam resource.
- RF-chain activity and pilot grouping are determined jointly by the matching outcome.

Resource Groups

Each AP-beam resource induces a UE group from the matching outcome:

$$\mathcal{G}_r = \{k : \mu_{k,r} = 1\}.$$

Gao

AP group: $\mathcal{G}_k = \{m : \mu(m, k) = 1\}$.

Risk is computed inside each AP group.

Beam Resource Matching

AP-beam resource group: $\mathcal{G}_r = \{k : \mu_{k,r} = 1\}$.

Risk is computed inside each resource group.

Key replacement

Group structure shifts from AP groups (Gao) to AP-beam resource groups (Beam Resource Matching).

Weighted Beam Overlap

Pairwise interference between UE u and UE v is measured by a weighted overlap score:

$$W_{u,v} = w_{\mathrm{aa}}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{\mathrm{ai}}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{\mathrm{ia}}|\mathcal{I}_u \cap \mathcal{A}_v|.$$

- $w_{\mathrm{aa}}|\mathcal{A}_u \cap \mathcal{A}_v|$: shared active resources; dominant interference term.
- $w_{\mathrm{ai}}|\mathcal{A}_u \cap \mathcal{I}_v|$: active for u, moderate for v; secondary term.
- $w_{\mathrm{ia}}|\mathcal{I}_u \cap \mathcal{A}_v|$: moderate for u, active for v; secondary term.

Weight propagation

Weights propagate into the overlap ratio, resource group risk, and pilot reuse cost—not merely a diagnostic label.

Weighted Beam-Overlap Ratio

Beam Resource Matching replaces Gao's common-serving-AP ratio with a beam-domain weighted counterpart:

$$M^{B}_{u,v} = \frac{W_{u,v}}{|\mathcal{A}_u|} = \frac{w_{\mathrm{aa}}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{\mathrm{ai}}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{\mathrm{ia}}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Unweighted baseline (Mussbah-style)

$$|\mathcal{A}_u \cap \mathcal{A}_v| + |\mathcal{A}_u \cap \mathcal{I}_v| + |\mathcal{I}_u \cap \mathcal{A}_v|.$$

All overlap types weighted equally.

Directed ratio

$$M^{B}_{u,v} \neq M^{B}_{v,u}$$

in general; denominators $|\mathcal{A}_u|$ and $|\mathcal{A}_v|$ typically differ.

Resource Group Risk

Resource group risk aggregates the weighted beam-overlap ratio over UE pairs:

$$S_r = \sum_{u \neq v,\ u,v \in \mathcal{G}_r} M^{B}_{u,v}.$$

- Groups are processed in descending order of S_r.
- Within a group, unused pilots are assigned first.
- If no unused pilot remains, the pilot minimizing the weighted reuse cost is selected:

$$C(u,p) = \sum_{i:p_i=p} \left(M^{B}_{u,i} + M^{B}_{i,u} \right), \qquad p_u^{\star} = \arg\min_{p} C(u,p).$$

Weighted Assignment Chain

$$
\begin{tikzcd}
w_{\mathrm{aa}},\, w_{\mathrm{ai}},\, w_{\mathrm{ia}} \arrow[r] & W_{u,v} \arrow[r] & M^{B}_{u,v} \arrow[d] \\
 & p^{\star}_{u} & S_r,\ C(u,p) \arrow[l]
\end{tikzcd}
$$

Assignment implication

A single weighted signal $(w_{\mathrm{aa}}, w_{\mathrm{ai}}, w_{\mathrm{ia}})$ drives both group priority and pilot reuse cost.

Adaptive Pilot Count

The pilot count matches the largest resource group:

$$\tau_{\mathrm{p}}^{\mathrm{BRM}} = \max_r |\mathcal{G}_r|.$$

The matching stage enforces

$$|\mathcal{G}_r| \leq Q_{\mathrm{res}} = \tau_{\mathrm{p}}^{\mathrm{design}},$$

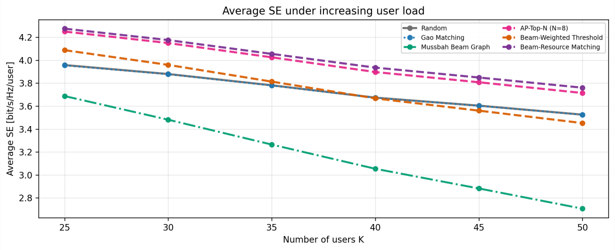
so the adaptive count is upper-bounded by the design budget:

$$\tau_{\mathrm{p}}^{\mathrm{BRM}} \leq \tau_{\mathrm{p}}^{\mathrm{design}}.$$

Pilot-count control

Mussbah's chromatic number grows unbounded with graph density. Beam Resource Matching caps the pilot count at the resource quota $Q_{\mathrm{res}} = \tau_{\mathrm{p}}^{\mathrm{design}}$.

Main Result: Moderate User Load



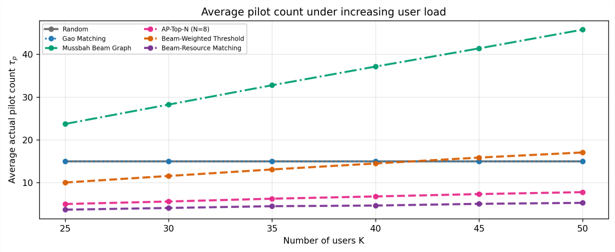
Evaluation configuration

- $L = 200$, $N = 8$
- $K = 25, \ldots, 50$
- $\tau_c = 150$
- $\tau_p^{\text{design}} = 15$
- 200 setups

Observed performance at $K = 50$

- Beam Resource Matching: $+6.66\%$ mean SE vs. Random.
- AP-Top-N: $+5.35\%$ mean SE vs. Random.
- Gao matches Random within noise.
- Mussbah underperforms due to inflated pilot count.

Actual Pilot Count Explains the Trend



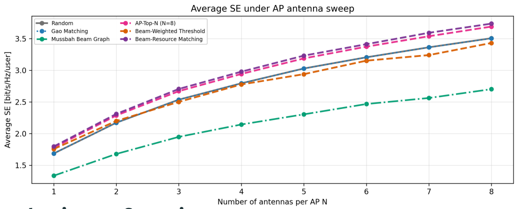
Pilot-overhead profile

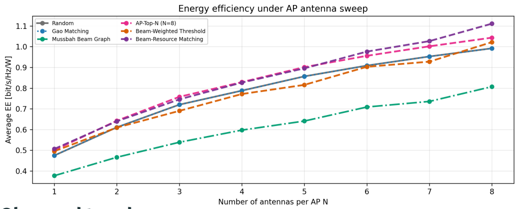
- Random and Gao consume the full design budget.
- Mussbah's chromatic number grows steeply with load.
- Beam Resource Matching holds τ_p below the design budget across all load levels.

Observed pilot count at $K = 50$

- Random / Gao: $\tau_\mathrm{p} = 15$.
- AP-Top-N: $\tau_\mathrm{p} \approx 7.79$.
- Beam Resource Matching: $\tau_\mathrm{p} \approx 5.29$.
- Mussbah: $\tau_\mathrm{p} \approx 45.76$.

Antenna Sweep: Robustness Across N





Evaluation configuration: $L = 200$, $K = 50$, $N = 1, \ldots, 8$, $\tau_{\mathrm{c}} = 150$, $\tau_{\mathrm{p}}^{\mathrm{design}} = 15$, 200 setups.

Observed trend

- Beam Resource Matching attains the highest mean SE across all tested N.
- At $N = 8$: $+6.57\%$ mean SE, $+12.01\%$ EE proxy.
- AP-Top-N provides a simpler AP-domain alternative with consistent positive gain.

Performance Mechanism of BRM

1. Beam-domain is more precise

$$\text{same AP} \neq \text{same beam}.$$

Two UEs can share an AP while occupying different beams.

2. Active beams are matched

- Active beams emerge from global UE-resource competition, not per-UE power alone.
- Resource quota Q_{res} bounds UE concentration per resource.

3. Conflict severity is weighted

$$M^{B}_{u,v} = \frac{w_{\mathrm{aa}}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{\mathrm{ai}}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{\mathrm{ia}}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Strong beam-domain collisions are prioritized during pilot assignment.

Pilot-Overhead Impact on SE

Spectral efficiency contains a pilot-overhead prelog factor:

$$\mathrm{SE}_k = \frac{\tau_{\mathrm{c}} - \tau_{\mathrm{p}}}{\tau_{\mathrm{c}}} \log_2(1 + \mathrm{SINR}_k).$$

- Reducing τ_{p} raises the prelog factor; SE increases provided SINR is preserved.
- A smaller active beam set reduces RF-chain activity.
- Combined effect lowers total active power and improves the EE proxy.

Design implication

Beam Resource Matching jointly reduces pilot overhead and RF-chain activity while preserving the dominant beam-domain interference structure.

Beam Resource Matching

Beam Resource Matching extends Gao's AP-domain many-to-many matching to Mussbah's beam-domain AP-beam resources.

- UEs are matched to resources $r = (l, n)$, each representing one AP-beam pair.
- The matching outcome defines the final active beam set:

$$\mathcal{A}_k = \{r : \mu_{k,r} = 1\}.$$

- The weighted ratio $M^B_{u,v}$ drives resource group risk S_r and pilot reuse cost $C(u, p)$.
- The actual pilot count is bounded by the resource quota:

$$\tau_\mathrm{p} = \max_r |\mathcal{G}_r| \leq \tau_\mathrm{p}^{\mathrm{design}}.$$

Core principle: prioritize dominant beam-domain collisions while bounding pilot overhead through resource quotas.